

1.1 SETS

Set is a well-defined collection of distinct objects. Sets are usually denoted by capital letters of English alphabet. Some examples are:

- (1) $A =$ Set of first six prime numbers (Descriptive form)
- (2) $B = \{1, 2, 3, 4, 5\}$ (Tabular form)
- (3) $C = \{x | x \in \mathbb{Z} \wedge -5 < x < 5\}$ (Set-builder form)

REVISION EXERCISE

1. Write the following sets in tabular and set builder forms.

- (i) $A =$ Set of first five natural numbers
- (ii) $B =$ Set of integers between -2 and 3 .
- (iii) $C =$ Set of prime numbers between 10 and 30 .

2. From the sets given below identify and write:

(i) Finite sets

(ii) Infinite sets

$$A = \{1, 2, 3, 4, 5\}, B = \{1, 3, 5, 7, \dots\}$$

$C =$ Set of composite numbers between 10 and 50 .

$D =$ Set of all multiples of 10 and $E = \{x | x \in \mathbb{Z} \wedge x < 0\}$

3. Define with examples:

- (i) Disjoint sets
- (ii) Overlapping sets
- (iii) Complement of a set

4. Verify with examples: (i) $A \cap A' = \emptyset$ (ii) $A \cup A' = \mathbf{U}$ (iii) $(A')' = A$

1.1.1 Recognize Some Important Sets:

Some important sets of numbers along with their notations are given as under:

- Set of Natural Numbers $\mathbb{N} = \{1, 2, 3, \dots\}$
- Set of Whole Numbers $\mathbb{W} = \{0, 1, 2, 3, \dots\}$
- Set of Integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ or $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \dots\}$
- Set of Even Integers $\mathbb{E} = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\}$ or $\mathbb{E} = \{0, \pm 2, \pm 4, \pm 6, \dots\}$
- Set of Odd Integers $\mathbb{O} = \{\dots, -5, -3, -1, 1, 3, 5, \dots\}$ or $\mathbb{O} = \{\pm 1, \pm 3, \pm 5, \dots\}$
- Set of Prime Numbers $\mathbb{P} = \{2, 3, 5, 7, \dots\}$
- Set of Composite Numbers $\mathbb{C} = \{4, 6, 8, 9, 10, 12, \dots\}$
- Set of Rational Numbers $\mathbb{Q} = \{x | x = \frac{p}{q}, p \wedge q \in \mathbb{Z}, q \neq 0\}$

1.1.2 Find a subset of a set

Subset: Set A is said to be a subset of a set B if every element of A is also an element of B. Symbolically, we write as $A \subseteq B$.

Example. Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5\}$ we see that $A \subseteq B$

Notes: (i) Empty set is a subset of every set $\emptyset \subseteq A$.

(ii) Every set is a subset of itself i.e. $A \subseteq A$.

1.1.3 Define Proper subset and an Improper subset

1. Proper subset

Let set A is a subset of set B but B is not a subset of A, then A is called proper subset of B, denoted by $A \subset B$.

Example 1. Let $A = \{1, 2, 3\}$ and $B = \{1, 2\}$

Here, $B \subseteq A$ but $A \not\subseteq B \Rightarrow B \subset A$.

Note: If B is not a proper subset of a set A, then we write it as $B \not\subset A$.

2. Improper subset

Two sets A and B are said to be improper subsets of each other if $A \subseteq B$ and $B \subseteq A$.

Note: There is no any particular symbol for an improper subset.

Example 2. Let $A = \{1, 2, 3, 4\}$ and $B = \{4, 3, 2, 1\}$.

Here A and B are improper subsets of each other.

Example 3. Find all possible subsets of $A = \{x, y, z\}$

All possible subsets are:

$$\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\}$$

Here Number of subsets $= 2^3 = 8$.

1.1.4. Find power set P(A) of a set A.

Power Set: Let A be a set. The set of all possible subsets of A is called power set of A. It is denoted by $P(A)$.

Example 4. Find the power set of $A = \{a, b\}$.

Solution: Here, number of subsets $= 2^2 = 4$. All possible subsets are: $\emptyset, \{a\}, \{b\}, \{a, b\}$. So, $P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.

Note: (i) Number of all possible subsets of an empty set is $1 = 2^0$

(ii) Number of all possible subsets of a set having one element is $2 = 2^1$

(iii) Number of all possible subsets of a set having two elements is $4 = 2^2$

(iv) Number of all possible subsets of a set having three elements is $8 = 2^3$

$\therefore$ Numbers of all possible subsets of a set having "n" elements is 2^n

EXERCISE 1.1**1. Indicate which one is a subset of the other**

- (i) $A = \{5, 6, 7\}$ and $B = \{1, 2, 3, \dots, 10\}$
(ii) $C = \{2\}$ and $D = \{2, 3, 5\}$
(iii) $T = \{1, 2, 3, \dots, 8\}$ and $S = \{2, 4, 6\}$

2. Find any three proper subsets of:

- (i) $A = \{a, e, i, o, u\}$ (ii) $B = \{x, y\}$

3. Find any two proper subsets and one improper subset of:

- (i) $X = \{2, 4\}$ (ii) $Y = \text{Set of prime numbers less than 4}$

4. Find all possible subsets of:

- (i) $F = \{1, 2, 3\}$ (ii) $A = \text{Set of even prime numbers}$
(iii) $B = \text{Set of odd prime numbers less than 7}$
(iv) $C = \{x/x \in \mathbb{Z} \wedge -2 < x < 2\}$ (v) $D = \{x/x \in \mathbb{E}^+ \wedge x \leq 6\}$

5. Find the power set of:

- (i) $A = \{1, 3, 5\}$ (ii) $B = \{a, b, c, d\}$ (iii) $C = \emptyset$

6. Find a set which has only one proper subset.**7. Find a set which has no proper subset at all.****8. Find improper subsets of the following:**

- (i) $A = \{x/x \in \mathbb{Q} \wedge x \leq 7\}$ (ii) $B = \{x/x \in \mathbb{P} \wedge x \leq 11\}$

9. Let $X = \{20, 30, 40, 50, 60, 80, 100\}$. Find the number of elements of the following subsets of X

- (i) $A = \text{Set of numbers divisible by 2.}$
(ii) $B = \text{Set of numbers divisible by 8.}$
(iii) $C = \text{Set of numbers divisible by 15.}$
(iv) $D = \text{Set of numbers divisible by 25.}$

10. Decide which one is a subset of the other.

- (i) $\mathbb{N}$ and $\mathbb{W}$ (ii) $\mathbb{Z}$ and $\mathbb{N}$ (iii) $\mathbb{P}$ and $\mathbb{N}$
(iv) $\mathbb{E}$ and $\mathbb{Z}$ (v) $\mathbb{Q}$ and $\mathbb{Q}$ (vi) $\mathbb{Q}$ and $\mathbb{E}$

1.2 OPERATIONS ON SETS

1.2.1 Verify commutative and associative laws with respect to union and intersection.

(A) Commutative laws w.r.t union and intersection.

Let A and B be any two sets.

Then commutative law w.r.t. union is:

$$A \cup B = B \cup A$$

And commutative law w.r.t. intersection is:

$$A \cap B = B \cap A$$

Example 1. If $A = \{1, 2, 3\}$ and $B = \{1, 3, 5\}$, then verify commutative laws w.r.t. union and intersection.

Solution: (i) Commutative law w.r.t union

We have to show that

$$A \cup B = B \cup A$$

$$\text{L.H.S.} = A \cup B$$

$$= \{1, 2, 3\} \cup \{1, 3, 5\}$$

$$= \{1, 2, 3, 5\}$$

$$\text{R.H.S.} = B \cup A$$

$$= \{1, 3, 5\} \cup \{1, 2, 3\}$$

$$= \{1, 2, 3, 5\}$$

$$\text{As L.H.S.} = \text{R.H.S.}$$

$$\therefore A \cup B = B \cup A.$$

Hence verified

Solution: (ii) Commutative law w.r.t. intersection

We have to show that

$$A \cap B = B \cap A$$

$$\text{L.H.S.} = A \cap B$$

$$= \{1, 2, 3\} \cap \{1, 3, 5\}$$

$$= \{1, 3\}$$

$$\text{R.H.S.} = B \cap A$$

$$= \{1, 3, 5\} \cap \{1, 2, 3\}$$

$$= \{1, 3\}$$

$$\text{As L.H.S.} = \text{R.H.S.}$$

$$\therefore A \cap B = B \cap A.$$

Hence verified

(B) Associative laws w.r.t. union and intersection.

Let A, B and C be any three sets,

then associative law w.r.t. union is:

$$A \cup (B \cup C) = (A \cup B) \cup C$$

And associative law w.r.t.

intersection is:

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Example 2. If $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ and $C = \{3, 4, 5\}$, then verify:

$$(i) A \cap (B \cap C) = (A \cap B) \cap C \quad (ii) A \cup (B \cup C) = (A \cup B) \cup C.$$

Solution:(i) $A \cap (B \cap C) = (A \cap B) \cap C$. [Associative law w.r.t. intersection]

$$\begin{array}{l|l} \text{L.H.S.} = A \cap (B \cap C) & \text{R.H.S.} = (A \cap B) \cap C \\ = \{1, 2, 3\} \cap (\{2, 3, 4\} \cap \{3, 4, 5\}) & = (\{1, 2, 3\} \cap \{2, 3, 4\}) \cap \{3, 4, 5\} \\ = \{1, 2, 3\} \cap \{3, 4\} = \{3\} & = \{2, 3\} \cap \{3, 4, 5\} = \{3\} \end{array}$$

As L.H.S. = R.H.S.

$\therefore A \cap (B \cap C) = (A \cap B) \cap C$. Hence verified.

Solution: (ii) $A \cup (B \cap C) = (A \cup B) \cap C$. [Associative Law w.r.t. to union]

$$\begin{array}{l|l} \text{L.H.S.} & \text{R.H.S.} \\ = A \cup (B \cap C) & = (A \cup B) \cap C \\ = \{1, 2, 3\} \cup (\{2, 3, 4\} \cap \{3, 4, 5\}) & = (\{1, 2, 3\} \cup \{2, 3, 4\}) \cap \{3, 4, 5\} \\ = \{1, 2, 3\} \cup \{2, 3, 4\} & = \{1, 2, 3, 4\} \cap \{3, 4, 5\} \\ = \{1, 2, 3, 4\} \cup \{3, 4, 5\} & = \{1, 2, 3, 4, 5\} \\ = \{1, 2, 3, 4, 5\} & \end{array}$$

As L.H.S. = R.H.S.

Hence verified.

$\therefore A \cup (B \cap C) = (A \cup B) \cap C$.

1.2.2 To verify distributive laws

Let A, B and C be any three sets. Then distributive law of union over intersection is:

And distributive law of

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

intersection over union is:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Example:

If $A = \{1, 2, 3\}$, $B = \{2, 4, 6\}$ and $C = \{2, 3, 6\}$, then verify distributive laws.

(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, (ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Solution: (i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$$\begin{aligned} \text{L.H.S.} &= A \cup (B \cap C) = \{1, 2, 3\} \cup (\{2, 4, 6\} \cap \{2, 3, 6\}) \\ &= \{1, 2, 3\} \cup \{2, 6\} = \{1, 2, 3, 6\}. \end{aligned}$$

$$\begin{aligned} \text{R.H.S.} &= (A \cup B) \cap (A \cup C) = (\{1, 2, 3\} \cup \{2, 4, 6\}) \cap (\{1, 2, 3\} \cup \{2, 3, 6\}) \\ &= \{1, 2, 3, 4, 6\} \cap \{1, 2, 3, 6\} = \{1, 2, 3, 6\}. \end{aligned}$$

As L.H.S. = R.H.S.

$\therefore A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. Hence verified.

Solution: (ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

$$\begin{aligned}\text{L.H.S.} &= A \cap (B \cup C) = \{1, 2, 3\} \cap (\{2, 4, 6\} \cup \{2, 3, 6\}) \\ &= \{1, 2, 3\} \cap \{2, 3, 4, 6\} = \{2, 3\}.\end{aligned}$$

$$\begin{aligned}\text{R.H.S.} &= (A \cap B) \cup (A \cap C) = (\{1, 2, 3\} \cap \{2, 4, 6\}) \cup (\{1, 2, 3\} \cap \{2, 3, 6\}) \\ &= \{2\} \cup \{2, 3\} = \{2, 3\}.\end{aligned}$$

$$\text{L.H.S.} = \text{R.H.S.}$$

$\therefore A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. Hence verified.

1.2.3 State and Verify De Morgan's Laws.

Let A and B be subsets of a universal set **U**.

Then De Morgan's Laws are:

$$(i) (A \cup B)' = A' \cap B', \quad (ii) (A \cap B)' = A' \cup B'.$$

Example. If $U = \{1, 2, 3, \dots, 6\}$, $A = \{1, 3, 5\}$

and $B = \{2, 4, 6\}$, verify any one of the De Morgan's Laws.

Solution: One of the De Morgan's Law is: $(A \cap B)' = A' \cup B'$

$$\text{L.H.S. Here } A \cap B = \{1, 3, 5\} \cap \{2, 4, 6\} = \{\}$$

$$\begin{aligned}\text{Therefore } (A \cap B)' &= U - (A \cap B) = \{1, 2, 3, \dots, 6\} - \{\} \\ &= \{1, 2, 3, \dots, 6\} = \text{L.H.S.}\end{aligned}$$

$$\text{R.H.S. Here } A' = U - A$$

$$= \{1, 2, 3, \dots, 6\} - \{1, 3, 5\} = \{2, 4, 6\}$$

$$B' = U - B = \{1, 2, 3, \dots, 6\} - \{2, 4, 6\} = \{1, 3, 5\}$$

$$\begin{aligned}\text{Therefore } A' \cup B' &= \{2, 4, 6\} \cup \{1, 3, 5\} \\ &= \{1, 2, 3, 4, 5, 6\} = \text{R.H.S.}\end{aligned}$$

$$\therefore (A \cap B)' = A' \cup B'. \quad \text{Hence verified.}$$

Activity: Let $U = \{a, b, c, d, e, f\}$, $A = \{a, b, c, d\}$ and $B = \{c, d, e, f\}$

Verify (i) $(A \cup B)' = A' \cap B'$

$$\text{Here } A \cup B = \dots\dots\dots$$

$$(A \cup B)' = \dots\dots\dots$$

$$A' = \dots\dots\dots, B' = \dots\dots\dots$$

$$A' \cap B' = \dots\dots\dots$$

$$\therefore \text{L.H.S.} = \dots\dots\dots$$

$$\text{R.H.S.} = \dots\dots\dots$$

$$\text{Thus, L.H.S.} = \dots\dots\dots$$

(ii) $(A \cap B)' = A' \cup B'$

$$\text{Here } A \cap B = \dots\dots\dots$$

$$(A \cap B)' = \dots\dots\dots$$

$$A' = \dots\dots\dots$$

$$B' = \dots\dots\dots$$

$$A' \cup B' = \dots\dots\dots$$

$$\therefore \text{L.H.S.} = \dots\dots\dots$$

$$\text{R.H.S.} = \dots\dots\dots$$

$$\text{Thus, } \dots\dots\dots = \text{R.H.S.}$$



August De Morgan (1806-1871) a British mathematician who formulated De Morgan Law.

EXERCISE 1.2

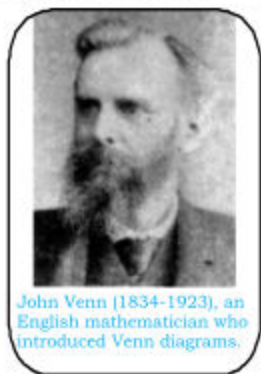
- If $A = \{1, 2, 3, 4, 5\}$, $B = \{1, 3, 5, 7\}$ and $C = \{2, 4, 6, 8\}$, then verify:
 - Commutative law w.r.t. intersection and union
 - Associative law w.r.t union and intersection
 - Distributive law of intersection over union
 - Distributive law of union over intersection.
- If $P = \{a, b, c, d, e\}$, $Q = \{a, e, i, o, u\}$ and $R = \{c, d, e, i\}$, then verify that
 - $P \cup Q = Q \cup P$
 - $P \cap Q = Q \cap P$
 - $P \cap (Q \cap R) = (P \cap Q) \cap R$
 - $P \cup (Q \cap R) = (P \cup Q) \cap (P \cup R)$
- If $U = \{1, 2, 3, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4, 6, 8, 10\}$, then verify that: (i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$
- If $P = \{x | x \in \mathbb{Q} \wedge x \leq 11\}$, $Q = \{x | x \in \mathbb{P} \wedge 3 \leq x \leq 11\}$ and $R = \{x | x \in \mathbb{E} \wedge x \leq 10\}$, then verify distributive laws.
- If $U = \{x | x \in \mathbb{N} \wedge x \leq 11\}$, $A = \{x | x \in \mathbb{E} \wedge 2 < x < 10\}$ and $B = \{x | x \in \mathbb{P} \wedge 3 \leq x \leq 11\}$, then verify De-Morgan's Laws.

1.3 VENN DIAGRAM

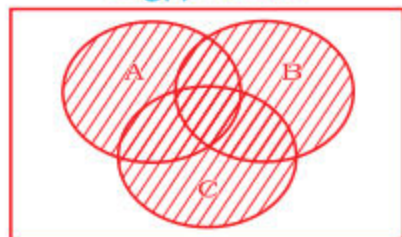
A diagram in which we represent sets pictorially as circles within an enclosing rectangle (universal set), common elements of the sets are represented by intersection of circles, termed as Venn diagram.

1.3.1 Demonstrate union and intersection of three overlapping sets through Venn Diagram

Here Union and Intersection of three overlapping sets is shown through Venn diagram in the following figures.

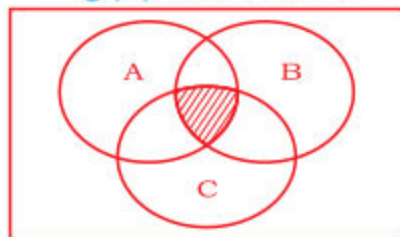


Fig(i): Union



Shaded portion represents Union

Fig (ii): Intersection



Shaded portion represents Intersection

1.3.2 To Verify Associative and Distributive Laws through Venn Diagram

Associative and distributive laws can easily be verified through Venn diagrams as explained in the following examples.

Example. If $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ and $C = \{3, 4, 5\}$, then verify through Venn diagrams.

$$(i) A \cap (B \cap C) = (A \cap B) \cap C, \quad (ii) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution: (i) $A \cap (B \cap C) = (A \cap B) \cap C$

$$\text{L.H.S.} = A \cap (B \cap C).$$

Here  shaded portion represents $B \cap C$.

According to Fig (i); $B \cap C = \{3, 4\}$

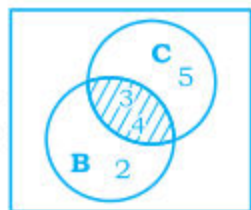


Figure (i)

According to Fig (ii); $A \cap (B \cap C) = \{3\}$

$$\text{R.H.S.} = (A \cap B) \cap C$$

Here  shaded portion represents $A \cap (B \cap C)$.

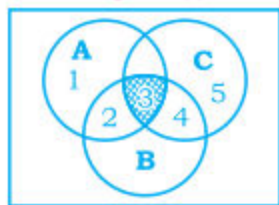


Figure (ii)

According to Fig (iii); $A \cap B = \{2, 3\}$

Here  shaded portion represents $(A \cap B)$.

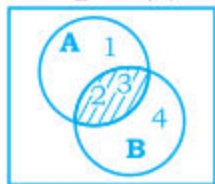


Figure (iii)

And according to Fig (iv); $(A \cap B) \cap C = \{3\}$

$$\text{Thus, L.H.S.} = \text{R.H.S.}$$

Here  shaded portion represents $(A \cap B) \cap C$.

$$\therefore A \cap (B \cap C) = (A \cap B) \cap C.$$

Hence verified, the associative law w.r.t. Intersection.

Activity: Verify the Associative Law w.r.t. union.

Solution: (ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$$\text{L.H.S.} = A \cup (B \cap C).$$

Here  shaded portion represents $B \cap C$.

According to Fig (v); $B \cap C = \{3, 4\}$.

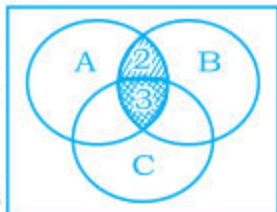


Figure (iv)

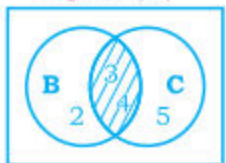
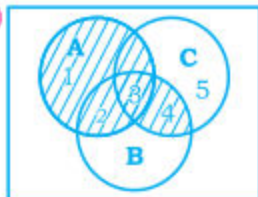


Fig (v)

According to Fig (vi); $A \cup (B \cap C) = \{1, 2, 3, 4\}$ Fig (vi)

R.H.S. = $(A \cup B) \cap (A \cup C)$

Here  and  shaded portion represents $A \cup (B \cap C)$.



According to Fig (vii); $A \cup B = \{1, 2, 3, 4\}$

Here  shaded portion represents $A \cup B$

Fig (vii)



According to Fig (viii); $A \cup C = \{1, 2, 3, 4, 5\}$.

Here  shaded portion represents $A \cup C$

Fig (viii)



According to Figure (ix) $(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4\}$.

Thus, L.H.S. = R.H.S.

Here  shaded portion represents $(A \cup B) \cap (A \cup C)$

$\therefore A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. Hence verified,
The Distributive Law of Union over Intersection.

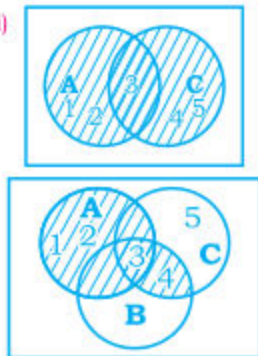


Fig (ix)

Activity: Students themselves verify Distributive Law of Intersection over union through Venn Diagram..

EXERCISE 1.3

- If $A = \{5, 6, 7\}$, $B = \{6, 7, 8\}$ and $C = \{7, 8, 9\}$, then represent:
(i) $A \cup (B \cap C)$ (ii) $A \cap (B \cap C)$ through Venn diagrams.
- If $A =$ Set of even natural numbers less than 15
 $B =$ Set of Prime numbers less than 15 and
 $C =$ Set of Odd natural numbers less than 15 then show the following by using Venn diagrams:
(i) $A \cup (B \cap C)$ (ii) $A \cap (B \cap C)$ (iii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- If $A = \{a, b, c\}$, $B = \{b, c, d\}$ and $C = \{c, d, e\}$, verify through Venn diagram that:
(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
(ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (iii) $A \cap (B \cap C) = (A \cap B) \cap C$
- If $D = \{x | x \in \mathbb{N} \wedge x \leq 10\}$, $E = \{x | x \in \mathbb{W} \wedge x \leq 8\}$
 $F = \{x | x \in \mathbb{P} \wedge x \leq 11\}$, then verify by using Venn diagram.
(i) $D \cup (E \cap F) = (D \cup E) \cap (D \cup F)$ (ii) $D \cap (E \cup F) = (D \cap E) \cup (D \cap F)$

REVIEW EXERCISE 1

1. Choose the correct answer.

(i) Which one is a subset of $\mathbb{W}$?

- (a)
- $\mathbb{Z}$
- (b)
- $\mathbb{N}$
- (c)
- $\mathbb{Q}$
- (d)
- $\mathbb{E}$

(ii) Number of all possible subsets of $A = \{a, e, i, o, u\}$ is _____.

- (a) 8 (b) 16 (c) 32 (d) 64

(iii) For any three sets A, B and C , $(A \cup B) \cap (A \cup C) =$ _____.

- (a)
- $A \cap (B \cup C)$
- (b)
- $A \cup (B \cup C)$
- (c)
- $A \cup (B \cap C)$
- (d) None of these

(iv) In Venn diagram, universal set is usually denoted by _____.

- (a) Triangle (b) Rectangle (c) Circle (d) Star

2. Find $P(A)$ if $A = \{x, y, z\}$.3. If $A = \{1, 2, 3\}$, $B = \{2, 4, 6\}$ and $C = \{x | x \in \mathbb{W} \wedge x \leq 4\}$ then verify:

- (i) $(A \cap B) \cap C = A \cap (B \cap C)$ (ii) $(A \cup B) \cup C = A \cup (B \cup C)$
 (iii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (iv) $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$
 (v) $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ (vi) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

4. If $U = \{1, 2, 3, \dots, 10\}$, $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8, 10\}$ then verify De Morgan's Laws.

5. Choose the correct answer.

(i) $A = \{2, 4, 6, 8\}$, $B = \{1, 3, 5, 7, 9\}$, then $A \cap B$ is:

- (a)
- $\{2, 4\}$
- (b)
- $\{\}$
- (c)
- $\{1, 3\}$
- (d)
- $\{2, 9\}$

(ii) $A = \{0, 2\}$, $B = \{1, 3\}$, then $A \cup B$ is:

- (a)
- $\{0, 2\}$
- (b)
- $\{1, 3\}$
- (c)
- $\{\}$
- (d)
- $\{0, 1, 2, 3\}$

(iii) If $P = \{0, 2, 6\}$ then improper subset of P is:

- (a)
- $\{0, 2\}$
- (b)
- $\{0, 6\}$
- (c)
- $\{0, 2, 6\}$
- (d)
- $\{2, 6\}$

6. Which of the following is a power set and write its original set.

- (i) $\emptyset$ (ii) $\{\emptyset, x\}$ (iii) $\{\emptyset, \{x\}, \{y\}\}$
 (iv) $\{\emptyset, \{x\}, \{y\}, \{x, y\}\}$ (v) $\{\emptyset, \{x\}, \{y\}, \{z\}\}$ (vi) $\{\emptyset\}$

SUMMARY

- Well-defined collection of distinct objects is called a set.
- Important sets: $\mathbb{N} = \{1, 2, 3, \dots\}$ $\mathbb{W} = \{0, 1, 2, 3, \dots\}$
 $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$ $\mathbb{Q} = \{x | x = \frac{p}{q}, p \wedge q \in \mathbb{Z}, q \neq 0\}$
 $\mathbb{E} = \{0, \pm 2, \pm 4, \dots\}$, $\mathbb{O} = \{\pm 1, \pm 3, \pm 5, \dots\}$, $\mathbb{P} = \{2, 3, 5, 7, \dots\}$
- If every element of set A is also an element of set B, then A is a subset of B.
- Any subset of set A is called a proper subset of A, if it is not equal to A.
- Any subset of set A is called an improper subset of A, if it is equal to A.
- The set of all possible subsets of set A is called power set of A. It is denoted as $P(A)$.
- Empty set is a subset of every set.
- If A consists of n elements then $P(A)$ consist of 2^n elements.
- Commutative laws: (i) $A \cup B = B \cup A$ (ii) $A \cap B = B \cap A$
- Associative laws: (i) $A \cup (B \cap C) = (A \cup B) \cap C$
 (ii) $A \cap (B \cup C) = (A \cap B) \cup C$
- Distributive laws: (i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 (ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- De Morgan's laws:
 (i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$
- A diagram in which we represent sets pictorially as circles within an enclosing rectangle (universal set), common elements of the sets are represented by intersection of circles, termed as Venn diagram.